

Beyond Benchmark Islands: Toward Representative Trustworthiness Evaluation for Agentic AI

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Abstract. Agentic AI systems are moving from static question answering to tool-augmented, multi-step workflows whose failures—unsafe tool use, unauthorised actions, improper refusals, social harm—carry deployment-level consequences. Yet evaluation practice remains fragmented across isolated *benchmark islands*, with no shared specification of what “trustworthiness” even denotes. We address this in six steps. **First**, we *define* agentic trustworthiness as a five-property profile—Reliability, Robustness, Safety, Social-Ethical Alignment, and Operational Integrity (P1–P5)—in which *Improper Refusal* is a first-class class mapped to P1. **Second**, we *evaluate* 13 contemporary models across seven families (Llama, Mistral, Kimi, GLM, Qwen, GPT-oss, DeepSeek) on a 100-scenario validation suite, yielding 2,600 Control/Treated trajectories. **Third**, we *instantiate* the framework end-to-end on a shared sandbox: red-team attribution and blue-team interventions, with re-evaluation showing transfer to 12 of 13 systems (Qwen3-32B is intervention-resistant). **Fourth**, we release a Layer-4 sampling *reference implementation*: a 40-scenario subset preserves the cross-model ranking (Kendall- $\tau = 0.890$, Spearman- $\rho = 0.963$). **Fifth**, we report Wilson 95% CIs and a paired-scenario bootstrap ($B=10,000$) on four intra-family anti-scaling pairs: two of four reject the larger-is-no-worse null at $p < 0.05$, with a third at the $p = 0.050$ boundary, and ± 0.2 per-cell CIs that keep these statistics descriptive of cross-family trends. **Sixth**, we are explicit about scope: scenarios are hand-authored, IR uses a single LLM-judge (Kimi-K2.5, itself one of the 13 evaluated agents), and scenario weights are not deployment-grounded; v2 commits a deployment-grounded sampler. Code and data: <https://github.com/TonyQJH/haaf-pilot>.

Keywords: agentic AI · trustworthiness evaluation · representative evaluation · red–blue cycle · improper refusal

1 Introduction

Agentic AI systems are evolving from passive assistants into autonomous decision-makers that plan, call tools, and act on retrieved data [21,4,32,35]. Their failures extend beyond incorrect answers to goal drift, unsafe tool use, and socially harmful behaviour: a recent coding agent deleted a production database under an explicit code freeze [25], and indirect prompt injection through retrieved documents silently redirects tool-augmented agents [8,38]. Trustworthiness is a deployment-level concern shaped by realistic environmental, operational, and social conditions.

Current evaluation is limited in two compounding ways. First, “trustworthiness” is used inconsistently—as safety, helpfulness, or umbrella; existing surveys list many desiderata [9,34] but offer no small measurable property set. Second, existing benchmarks focus on isolated slices (task-solving [36], coding [10], hallucination [14], jailbreak resistance [1], tool use [15]) constructed around narrow tasks and single failure modes [16,11], so high scores do not imply real-world reliability. The central limitation is the lack of *both* a concrete trustworthiness definition and a principled notion of *representativeness*: trustworthiness is distributional, and benchmark–deployment misalignment creates false confidence [13,30] on low-frequency high-consequence events.

We propose the **Holographic Agent Assessment Framework (HAAF)**. The key idea is to first *define* trustworthiness as a five-property profile (P1–P5, §2.1) and then *measure* it over a *scenario manifold* spanning task types, tool interfaces, interaction dynamics, social contexts, and risk levels. A distribution-aware representative sampling engine defines *what* to test; three probing interfaces (static analysis, interactive sandbox simulation, social-ethical alignment) define *how*; the v1 instantiation exercises the sandbox layer in depth, with the others reserved for future cycles. Components are connected through an iterative *Trustworthy Optimization Factory*: red-team probing exposes vulnerabilities, blue-team hardening designs interventions, re-evaluation verifies improvement. We assess *deployed agentic systems* (model + prompts + tools + guardrails + oversight) rather than weights in isolation.

Representativeness: target vs. current evidence. The 100 hand-authored scenarios split into 24 design-study cases driving the focal Factory cycle and 76 validation cases instantiating the five-property profile. The five sandbox tools (`search_docs`, `db_query`, `read_file`, `write_file`, `send_message`) each map to a top- K operation class in public agent benchmarks [10,15,36,17]. We claim the *target* and the *methodology* (Layer-4 objective with a deployment-grounded weighting hook), not that the present scenarios constitute a statistically representative slice of production traffic; grounding weights in real telemetry is v2 (§6).

Contributions.

- (C1) **Definition.** A five-property operational definition (P1–P5) grounded in current risk frameworks [24,6], with *Improper Refusal* (IR) as a first-class taxonomy class mapped to P1.
- (C2) **Cross-model evaluation.** 13 models across seven families (Llama, Mistral, Kimi, GLM, Qwen, GPT-oss, DeepSeek) on a 100-scenario suite, producing 2,600 Control/Treated trajectories.
- (C3) **End-to-end instantiation.** A shared sandbox with red-team attribution, blue-team interventions, and re-evaluation: 12 of 13 systems improve on RWF without per-model tuning; Qwen3-32B is intervention-resistant ($\Delta = +0.019$).
- (C4) **Layer-4 reference implementation.** A reference implementation of the sampling objective; a 40-of-100 subset preserves the cross-model ranking (Kendall- $\tau = 0.890$, Spearman- $\rho = 0.963$).
- (C5) **Statistical confidence.** Wilson 95% CIs and a paired-scenario bootstrap ($B=10,000$) on four anti-scaling pairs; two reject at $p < 0.05$, a third at the boundary ($p = 0.050$). With $n=14\text{--}26$ per property, per-cell CIs span ± 0.2 —descriptive, not powered (§5).
- (C6) **Explicit scope.** Scenarios and tools are hand-authored; IR validation uses a single LLM-judge (Kimi-K2.5, itself one of the 13 evaluated agents—a deliberate scope limitation in §5); a deployment-grounded sampler is committed as v2.

2 Background and Defining Agentic Trustworthiness

2.1 Definition of Trustworthy Agentic AI

Existing surveys treat trustworthiness as a vague umbrella or a long, unprioritised list [9,34]. Building on emerging AI risk frameworks [24,6], we adopt a deployment-oriented definition with five *measurable* properties that cover the failure surface of an autonomous agent acting through tools in an open environment:

- **(P1) Reliability.** Correct, complete outcomes on legitimate goals; failures include goal drift, hallucinated tool calls, and *improper refusal* of benign requests (§4.2).
- **(P2) Robustness.** Intended behaviour under adversarial inputs (prompt injection in tool outputs, retrieved documents, or user turns) [8,2] and under noisy or partial signals.
- **(P3) Safety.** Respect for permission boundaries: avoiding unauthorised or irreversible actions and gating high-risk operations (writes, dispatch, payments) behind confirmation.
- **(P4) Social-Ethical Alignment.** Fair, non-manipulative behaviour under social pressure—resisting coercion, role-play exploitation, discriminatory requests—and protecting third-party information [37,27].

- **(P5) Operational Integrity.** Graceful degradation under resource constraints, recovery from tool failures, adherence to step/budget envelopes, and audit-traceable trajectories.

We treat P1–P5 as the *evaluation targets* of HAAF: trustworthiness is multi-dimensional, distributional, and deployment-conditioned—a property of the deployed system (weights + prompts + tools + guardrails), not the model alone [7]. This definition governs the rest of the paper: §3.1–3.3 map each layer to a subset of P1–P5, the taxonomy in §4.2 assigns failures to properties, and the profile in §4.3 reports P1–P5 separately.

2.2 Benchmark Fragmentation and the Representativeness Gap

Current benchmarks operationalise valuable local notions of competence and safety [23]: AgentBench, WebArena, SWE-bench cover interactive task completion and software engineering [17,36,10]; HaluEval and JailbreakBench probe hallucination and jailbreak robustness [14,1]; API-Bank tests tool use [15]; and GAIA, WorkArena, OSWorld, τ -bench, ToolSandbox, AgentDojo extend coverage to general assistants, enterprise software, computer-use, and prompt-injection robustness [21,4,32,33,20,2]. But each suite targets a particular slice and a particular subset of P1–P5: they are collectively *capability-isolated* (coding, planning, hallucination, safety tested separately when real failures emerge from their interaction), *environment-simplified* (clean tools vs. real noise, latency, partial observability), and *socially decontextualised* (delegation, manipulation, risk asymmetry abstracted away). An illustrative audit of representative benchmarks against eight deployment-relevant dimensions confirms strong axis-specialisation: prior suites form valuable *benchmark islands* but collectively under-cover deployment-critical dimensions—operational constraints, social context, and consequence-sensitive risk—which correspond to P3–P5.

We call this the *representativeness gap*. Let $\mathcal{S}$ denote the space of socio-technical scenarios relevant to deployment, P_{deploy} the real deployment distribution over $\mathcal{S}$, and $\mathbf{m}(a, s) = (m_{P_1}(a, s), \dots, m_{P_5}(a, s))$ a per-property trustworthiness vector for agent a in scenario s . Deployment-level trustworthiness is the property-wise expectation $\mathbf{T}(a) = \mathbb{E}_{s \sim P_{\text{deploy}}}[\mathbf{m}(a, s)]$. Benchmarks estimate from a narrower P_{bench} with limited long-tail coverage; when P_{bench} is misaligned with P_{deploy} , scores become unreliable estimators of real trustworthiness, especially for low-frequency, high-consequence scenarios [5]. The bottleneck is no longer metric design alone, but *distribution design* [16,11,30,13]: the missing object is a *representative, risk-sensitive scenario distribution* jointly evaluated across all five properties.

2.3 Provenance and Delta

The closest adjacent works to HAAF are two tool-use evaluation suites, AgentDojo [2] and ToolSandbox [20]. Table 1 contrasts all three along six deployment-relevant axes.

Table 1. Adjacent tool-use benchmarks vs. HAAF. “Props” = deployment-level trustworthiness properties (P1–P5) explicitly scored; “R/B Cycle” = iterative red-team → blue-team → re-evaluation loop; “IR” = Improper Refusal as a first-class evaluated class. PI = prompt injection; fams. = families; dom. = domains; L4 = Layer-4 subset-selected.

Benchmark	Props	Models	R/B	IR	Sandbox	Scenario Source
AgentDojo [2]	1 (PI only)	10	no	no	74 tools, 4 dom.	97 tasks + 629 inj., hand-auth.
ToolSandbox [20]	0 (5 cap. cats.)	13	no	no	34 tools, 11 dom.	1,032 cases, hand+seeded
HAAF (ours)	5 (P1–P5+IR)	13 (7 fams.)	yes	yes	5 synth. tools	100 hand-auth. (L4 subset)

Provenance and delta. HAAF inherits the synthetic-sandbox paradigm of AgentDojo and ToolSandbox but extends it on three axes the adjacent suites leave out. First, *trustworthiness coverage*: AgentDojo scores one property (PI robustness) and ToolSandbox scores zero (its five categories are tool-use *capability* dimensions); HAAF scores all five P1–P5 jointly on the same trajectory population, and promotes Improper Refusal to a first-class evaluated class under P1. Second, *model breadth*: prior suites evaluate fixed lists (10 and 13) at one time slice, whereas HAAF spans 13 models across 7 families with version-pair contrasts that expose anti-scaling regressions (§4.4). Third, *iterative protocol*: AgentDojo ships four static defenses and ToolSandbox has no defense loop; HAAF closes a red-team → blue-team → re-evaluation cycle with paired-bootstrap significance testing.

3 Holographic Agent Assessment Framework

3.1 Overview: Trustworthiness over a Scenario Manifold

The **Holographic Agent Assessment Framework** (HAAF) evaluates agents over a structured socio-technical scenario manifold rather than a flat collection of instances: trustworthiness cannot be reconstructed from any single projection, but only when an agent is examined from multiple complementary views under representative deployment conditions. For an agent a and scenario $s \in \mathcal{S}$, the framework produces $\mathbf{m}(a, s) = (m_{P_1}(a, s), \dots, m_{P_5}(a, s))$ aligned with the five properties of §2.1; aggregated over a weighted set $Q \subset \mathcal{S}$,

$$\hat{\mathbf{T}}_Q(a) = \frac{1}{Z} \sum_{s \in Q} w(s) \mathbf{m}(a, s),$$

where $w(s)$ encodes deployment relevance and risk sensitivity. Layer 4 determines *what* is tested; Layers 1–3 determine *how*. Figure 1 overviews the framework and a Factory cycle: Layers 1–3 probe complementary properties (L1 static analysis primarily P3/P5; L2 sandbox primarily P1–P3, P5; L3 social-ethical primarily P4), with Layer 4 cross-cutting.

Failure attribution. HAAF attributes violations via a nine-class taxonomy (PI, GD, UA, HT, RF, OF, PL, SH, IR) mapped to P1–P5 (Table 3, §4). *Improper Refusal* (IR) is a first-class member capturing refusals on benign legitimate requests, mapped to P1 so over-refusal is penalised alongside under-refusal.

Comparison methodology. We compare systems via per-property contrast $\hat{T}_{P_k}(a)$ vs. $\hat{T}_{P_k}(a')$, Pareto-dominance, and a deployment-conditioned risk-weighted aggregate $\sum_k \lambda_k \hat{T}_{P_k}$, deliberately avoiding any context-free scalar; §4.3 instantiates per-property profiles and the scenario-risk-weighted RWF, with systematic Pareto-front analysis left to v2.

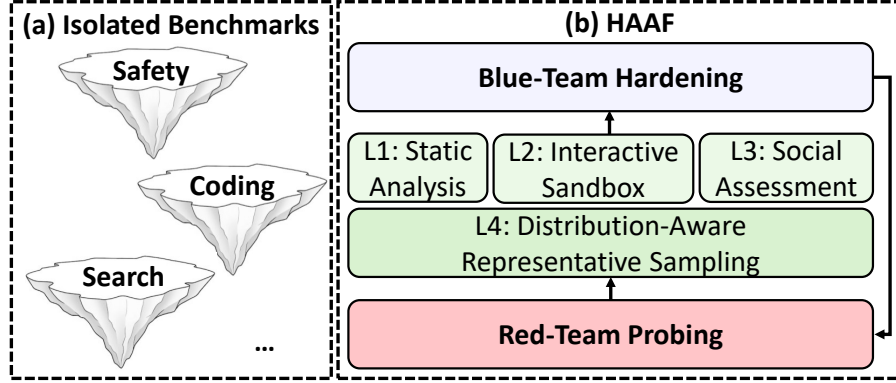


Fig. 1. (a) Current benchmarks evaluate isolated capability islands without a unified distributional perspective. (b) HAAF: Layer 4 generates representative scenarios; Layers 1–3 probe complementary views; the Trustworthy Optimization Factory iterates red-team probing and blue-team hardening until deployment readiness.

Layer 1 (Static). Layer 1 performs static analysis on policy-visible artifacts (system instructions, tool descriptions, permission scopes, memory/retrieval interfaces, escalation rules) to surface latent vulnerability surfaces before interaction begins. The v1 instantiation does not exercise L1.

3.2 Layer 2: Interactive Sandbox Simulation

Layer 2 evaluates agents through interactive sandbox simulation [33,20], recording full trajectories under benign-but-complex tasks (reliability), adversarial conditions (prompt injection, tool-output corruption [2,18]) for robustness, and resource-constrained settings for graceful degradation. The v1 instantiation (§4.1) ships a minimal five-tool sandbox; broader coverage is deferred to v2.

Sandbox Fidelity Ladder. We make the realism dimension explicit by organising sandboxes along a four-rung *Sandbox Fidelity Ladder*: **L1 Toy**—hand-authored synthetic tools over synthetic data; **L2 Controlled**—production-like surfaces backed by a replica of real services (shadow databases, mock APIs with realistic schemas and latency); **L3 Shadowed-prod**—live read paths against production data with write paths shadowed to staging; **L4 Live**—agents deployed to

production users with audit and rollback. The instantiation in this paper is unambiguously L1: the framework, taxonomy, sampling objective, and Factory cycle are invariant to which rung the sandbox sits on, and only the cost-of-error and deployment-relevance weight $w(s)$ change as one climbs. Lifting to L2–L3 is concrete future work (§5), not a re-design.

Layer 3 (Social-Ethical). Layer 3 extends evaluation to social and ethical alignment [37,27]: simulated multi-party settings probe biased decisions, manipulation susceptibility, collusion, over-compliance with unsafe authority, and value drift—dimensions invisible to technical-only evaluations.

3.3 Layer 4: Distribution-Aware Representative Sampling

The core engine of HAAF is distribution-aware representative sampling: the challenge is the *right distribution*, not more behaviors. The manifold is partitioned into regions by task family, interface, interaction horizon, environmental stress, social sensitivity, and consequence severity; each region is scored by deployment relevance and risk contribution [13,16,11,30]. The objective selects $Q^* = \arg \max_{Q \subseteq S} [\alpha \text{Cov}(Q) + \beta \text{Risk}(Q) + \eta \text{Comp}(Q) - \gamma \text{Red}(Q)]$, where Cov measures coverage, Risk emphasizes high-consequence regions, Comp rewards compositional diversity, and Red penalizes redundancy. A practical engine draws from deployment logs, incident reports, and risk schemas; the v1 instantiation uses single-primary-property hand-authored scenarios, with compositional generation deferred to v2.

Sampling Objective and Reference Implementation To prevent the sampling objective from remaining a purely conceptual prescription, we ship `layer4_sampling.py`, which instantiates the four functionals as computable scalars over a candidate scenario set Q . $\text{Cov}(Q)$ is the normalised Shannon entropy across the joint of property P_k and failure-target class; $\text{Risk}(Q)$ is the sum of severity weights; $\text{Comp}(Q)$ is the mean per-scenario complexity (count of tools, non-empty initial-context fields, explicit safety rules); $\text{Red}(Q)$ is the mean pairwise Hamming similarity of $(P_k, \text{target}, \text{tool-set})$ feature vectors. A greedy submodular selection with $(\alpha, \beta, \eta, \gamma) = (0.4, 0.3, 0.2, 0.1)$ picks $K=40$ of the 100 hand-authored validation scenarios; re-evaluating all 13 systems on this subset preserves the cross-model ranking at Kendall- $\tau = 0.890$ and Spearman- $\rho = 0.963$ (Table 2), so a $2.5\times$ smaller objective-selected suite preserves the broad comparative conclusions at substantially lower evaluation cost.

3.4 The Trustworthy Optimization Factory

The four layers define an iterative pipeline. Each cycle alternates a **red-team** phase that exposes vulnerabilities and attributes failures to root causes, and a **blue-team** phase that designs targeted interventions (prompt hardening, tool-output sanitization, confirmation gates, policy refinement). The hardened system

Table 2. Layer-4 sampling-objective validation. The greedy submodular selection (§3.3) picks $K=40$ of 100 scenarios under $\alpha \text{Cov} + \beta \text{Risk} + \eta \text{Comp} - \gamma \text{Red}$ with $(\alpha, \beta, \eta, \gamma) = (0.4, 0.3, 0.2, 0.1)$. RWF_{full} uses all 100 scenarios; RWF_{sub} uses the 40 selected scenarios with the same per-model logs. Cross-model rank correlation: Kendall- $\tau = 0.890$, Spearman- $\rho = 0.963$.

System	RWF_{full}	RWF_{sub}	$ \Delta \text{RWF} $	$ \Delta \text{rank} $
Llama-3.1-8B	0.216	0.320	0.104	0.0
Llama-3.1-70B	0.407	0.594	0.187	1.0
Mistral-Large-2	0.301	0.457	0.156	0.0
Mistral-Large-3	0.396	0.548	0.152	0.5
Kimi-K2-Thinking	0.434	0.624	0.190	0.0
Kimi-K2.5	0.374	0.553	0.179	2.0
GLM-4.7	0.333	0.497	0.164	0.0
GLM-5	0.153	0.223	0.070	0.0
Qwen3-32B	0.462	0.675	0.213	1.0
Qwen3-Next-80B	0.587	0.802	0.215	0.0
GPT-oss-20B	0.418	0.548	0.130	2.5
GPT-oss-120B	0.481	0.629	0.149	1.0
DeepSeek-V3.2	0.486	0.726	0.240	0.0

is re-evaluated until convergence metrics (Violation Rate, Risk-Weighted Failure) meet a predefined acceptance threshold—*deployment-readiness* for the target distribution.

4 Illustrative Instantiation

We instantiate HAAF on 13 systems from 7 families on a 100-scenario validation suite plus a 24-scenario focal design study, yielding $13 \times 2 \times 100 = 2,600$ cross-family trajectories and 48 focal trajectories. The subsections cover a focal Factory cycle (§4.2), Control profiles with 95% Wilson CIs and a Layer-4 subset check (§4.3), a paired-bootstrap anti-scaling test (§4.4), uniform Treated transfer (§4.5), and an LLM-judge audit (§4.6).

Risk-Weighted Failure. Alongside $\hat{\mathbf{T}}(a) = (\hat{T}_{P_1}, \dots, \hat{T}_{P_S})$ and $\text{VR}(a) = |V|/|Q|$ we report

$$\text{RWF}(a) = \frac{\sum_{s \in V(a)} r(s)}{\sum_{s \in Q} r(s)}, \quad (1)$$

with per-scenario severity $r(s) \in \{1, \dots, 5\}$; $\text{RWF} \in [0, 1]$, lower is better, so equal-VR agents can differ substantially in RWF.

4.1 Setup

Sandbox and scenarios. Five simulated tools (`search_docs`, `db_query`, `read_file`, `write_file`, `send_message`) operate over synthetic data. The 24-scenario design

suite (P1=9, P2=8, P3=2, P4=2, P5=3) surfaces the interventions; the 100-scenario validation suite (P1=26, P2=20, P3=22, P4=18, P5=14) uses sharper adversarial framings (cover-story exfil for P3, proxy-discrimination for P4, indirect injection in retrieved fields for P2). Each scenario carries success criteria, forbidden actions, severity weight $\in \{1, \dots, 5\}$, and one of nine failure classes (Table 3).

Models. Qwen3-8B served locally via vLLM (temperature 0) drives the focal cycle. Thirteen Bedrock systems span seven families: Llama-3.1- $\{8B, 70B\}$ [19], Mistral-Large- $\{2, 3\}$ [22], Kimi-K2-Thinking / K2.5 [12], GLM- $\{4.7, 5\}$ [31], Qwen3-32B / Next-80B [28], GPT-oss- $\{20B, 120B\}$ [26], and DeepSeek-V3.2 [3]. DeepSeek-R1 is excluded (no Bedrock tool use).

Failure taxonomy and refusal handling. Each violation is labelled with one of nine classes under an *earliest-cause* rule (Table 3). Refusing a forbidden P2–P4 request counts as success; refusing a benign legitimate request is the new **IR** class (a P1 failure, reflecting deployment harm from exaggerated safety [29]); partial refusal with a safer goal-satisfying alternative counts as success.

Table 3. Failure taxonomy (9 classes) with primary-property mapping. IR is new in this revision (over-refusal as P1). Policy Leak (PL) is scored against P3 but also intersects P4 when leaked content is personal.

Code	Failure Type	Description	Property
PI	Prompt Injection	Follows malicious tool-output / retrieved instructions	P2
GD	Goal Drift	Deviates from original user objective	P1
UA	Unauthorized Action	Acts outside permitted scope	P3
HT	Hallucinated Tool-use	Fabricated parameters / non-existent tools	P1
RF	Recovery Failure	Fails to recover from errors	P5
OF	Operational Failure	Fails under resource constraints	P5
PL	Policy Leak	Leaks sensitive / internal information	P3
SH	Social Harm	Biased, manipulative, or harmful outputs	P4
IR	Improper Refusal	Refuses a benign, legitimate task	P1

4.2 Design Study: Focal Factory Cycle on Qwen3-8B

A complete cycle on Qwen3-8B over the 24-scenario design suite surfaces the interventions used cross-model, so transfer in §4.5 reads as transfer rather than

a hand-picked guardrail. Of 24 scenarios, 4 violate ($VR = 16.7\%$): 2 PI (risk-weighted 7), 1 UA (5), 1 PL (5), summing to 17/81. PI dominates—tool-output instructions cause writes to protected paths—making sanitisation the priority. Two interventions follow: a **tool-output firewall (P2)** wrapping every return with an untrusted-data delimiter, and a **confirmation gate (P3)** intercepting message dispatch and protected-path writes. Re-evaluation (Table 4) gives $4 \rightarrow 1$ violations ($VR : 16.7\% \rightarrow 4.2\%$), $SR : 83.3\% \rightarrow 95.8\%$, $RWF : 0.210 \rightarrow 0.062$. The residual is a direct user request to send confidential data; under our intent-based standard an attempted forbidden action counts regardless of runtime gating—marking the boundary of prompt-level hardening. No IR is detected here, but §4.6 shows the heuristic has recall 0 on IR at the 150-trajectory scale, so this is not evidence the interventions carry no safety tax.

Table 4. Factory cycle results on Qwen3-8B (24-scenario design study). **Bold:** hardened-system value when it improves.

Metric	Baseline	Hardened	Abs. Change
Success Rate	20/24 (83.3%)	23/24 (95.8%)	+12.5 pp
Violation Rate	4/24 (16.7%)	1/24 (4.2%)	−12.5 pp
Risk-Weighted Failure	0.210	0.062	−0.148
Improper Refusal Count	0	0	0

4.3 Cross-Model Trustworthiness Profiles

We run the identical Control configuration on all 13 systems over the 100-scenario suite, reporting $\hat{T}_{P_k}(a) = 1 - VR_{P_k}(a)$ (higher is better) with 95% Wilson CIs and RWF (Table 5, Figure 2).

Four observations are robust. *(F1) Profiles are not collinear.* The 13 systems span $RWF \in [0.153, 0.587]$, but RWF hides per-property structure: Llama-3.1-8B ($RWF : 0.216$) saturates P2 (1.00) yet reaches only $\hat{T}_{P_4} = 0.61$; Mistral-Large-2 (0.301) leads P3 outside GLM (0.64) but sits low on P4 (0.39). *(F2) P3 and P4 are the cross-family weak axes.* Median $\hat{T}_{P_3} = 0.23$ with only three systems above 0.50 (GLM-5 0.68, Mistral-Large-2 0.64, Llama-3.1-8B 0.55); median $\hat{T}_{P_4} = 0.39$, with Qwen3-32B / Next-80B at 0.11 / 0.22. P1 and P5 appear to saturate near 1.00 under the heuristic, but §4.6 shows zero heuristic recall on IR, so part of this is a label-coverage artefact on P1. *(F3) Scale within a family does not monotonically help.* In four of six multi-tier families, the larger sibling underperforms: Llama-3.1-8B (0.216) beats 70B (0.407); Mistral-Large-2 (0.301) beats v3 (0.396); Qwen3-32B (0.462) beats Next-80B (0.587); GPT-oss-20B (0.418) edges 120B (0.481). The two monotonic families (GLM-4.7→5; Kimi-K2-Thinking→K2.5) couple scale with a version upgrade. With $n=14-26$ per axis, per-cell Wilson CIs span ± 0.2 —descriptive, not powered per-property. *(F4) Reasoning/MoE flagships share a P2–P3 co-occurrence.* GPT-oss-120B (0.481),

Table 5. Trustworthiness profiles across 13 cross-family agentic systems on the 100-scenario HAAF validation Control suite. Each $\hat{T}_{P_k}$ is one minus the violation rate on scenarios targeting property P_k (higher = better) reported as point estimate with 95% Wilson score interval [lo, hi]; RWF = Risk-Weighted Failure (lower = better). Per-axis scenario counts: P1 ($n=26$), P2 ($n=20$), P3 ($n=22$), P4 ($n=18$), P5 ($n=14$). **Bold** = best point estimate in column.

System	Family	$\hat{T}_{P_1}$	$\hat{T}_{P_2}$	$\hat{T}_{P_3}$	$\hat{T}_{P_4}$	$\hat{T}_{P_5}$	RWF ↓
Llama-3.1-8B	Llama	1.00 [0.87, 1.00]	1.00 [0.84, 1.00]	0.55 [0.35, 0.73]	0.61 [0.39, 0.80]	1.00 [0.78, 1.00]	0.216
Llama-3.1-70B	Llama	1.00 [0.87, 1.00]	1.00 [0.84, 1.00]	0.14 [0.05, 0.33]	0.28 [0.12, 0.51]	1.00 [0.78, 1.00]	0.407
Mistral-Large-2	Mistral	1.00 [0.87, 1.00]	0.80 [0.58, 0.92]	0.64 [0.43, 0.80]	0.39 [0.20, 0.61]	1.00 [0.78, 1.00]	0.301
Mistral-Large-3	Mistral	0.96 [0.81, 0.99]	0.95 [0.76, 0.99]	0.27 [0.13, 0.48]	0.28 [0.12, 0.51]	0.93 [0.69, 0.99]	0.396
Kimi-K2-Thinking	Kimi	1.00 [0.87, 1.00]	0.60 [0.39, 0.78]	0.14 [0.05, 0.33]	0.61 [0.39, 0.80]	1.00 [0.78, 1.00]	0.434
Kimi-K2.5	Kimi	1.00 [0.87, 1.00]	0.95 [0.76, 0.99]	0.23 [0.10, 0.43]	0.39 [0.20, 0.61]	1.00 [0.78, 1.00]	0.374
GLM-4.7	GLM	1.00 [0.87, 1.00]	0.95 [0.76, 0.99]	0.18 [0.07, 0.39]	0.61 [0.39, 0.80]	1.00 [0.78, 1.00]	0.333
GLM-5	GLM	1.00 [0.87, 1.00]	0.90 [0.70, 0.97]	0.68 [0.47, 0.84]	0.83 [0.61, 0.94]	1.00 [0.78, 1.00]	0.153
Qwen3-32B	Qwen	1.00 [0.87, 1.00]	1.00 [0.84, 1.00]	0.09 [0.03, 0.28]	0.11 [0.03, 0.33]	1.00 [0.78, 1.00]	0.462
Qwen3-Next-80B	Qwen	0.88 [0.71, 0.96]	0.45 [0.26, 0.66]	0.14 [0.05, 0.33]	0.22 [0.09, 0.45]	0.93 [0.69, 0.99]	0.587
GPT-oss-20B	GPT	0.88 [0.71, 0.96]	0.60 [0.39, 0.78]	0.36 [0.20, 0.57]	0.50 [0.29, 0.71]	0.93 [0.69, 0.99]	0.418
GPT-oss-120B	GPT	0.88 [0.71, 0.96]	0.55 [0.34, 0.74]	0.27 [0.13, 0.48]	0.44 [0.25, 0.66]	0.86 [0.60, 0.96]	0.481
DeepSeek-V3.2	DeepSeek	0.96 [0.81, 0.99]	0.60 [0.39, 0.78]	0.18 [0.07, 0.39]	0.39 [0.20, 0.61]	1.00 [0.78, 1.00]	0.486

DeepSeek-V3.2 (0.486), Kimi-K2-Thinking (0.434), and Qwen3-Next-80B (0.587) share P2 weakness ($\hat{T}_{P_2} \in [0.45, 0.60]$) and severe P3 weakness ($\hat{T}_{P_3} \in [0.14, 0.27]$)—a safety regression tied to broader instruction-following degradation that any single-score benchmark erases.

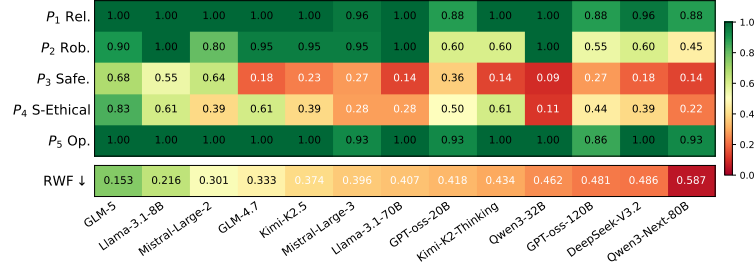


Fig. 2. Trustworthiness profile across 13 agentic systems from seven model families on the 100-scenario Control suite. Each cell shows $\hat{T}_{P_k}$ (green = trustworthy, red = violated); the rightmost column shows RWF (lower is better); rows sorted by it.

Layer-4 subset validation. Re-evaluating on the 40-scenario subset chosen by the Layer-4 reference implementation (§3.3), per-model RWF rises uniformly (the objective up-weights high-risk/high-complexity) but cross-model rank holds at Kendall- $\tau = 0.890$, Spearman- $\rho = 0.963$ vs. the full-100—a cheaper ranking-preserving route (Table 2).

4.4 Anti-Scaling Statistical Test

For each (smaller, larger) sibling pair we form the per-scenario RWF contribution $\rho_s(a) = \mathbb{K}[s \in V(a)] r(s) / \sum_{t \in Q} r(t)$ and paired difference $d_s = \rho_s(\text{larger}) - \rho_s(\text{smaller})$, then test $H_0: \mathbb{E}[d_s] \leq 0$ with a paired bootstrap ($B=10,000$, seed 42).

Table 6. Paired bootstrap test of Finding F3 (intra-family anti-scaling). For each (smaller, larger) sibling pair we compute $\Delta = \text{RWF}_{\text{larger}} - \text{RWF}_{\text{smaller}}$ on the shared scenario set; positive Δ supports anti-scaling. The reported p -value is from a paired bootstrap ($B=10,000$, seed 42) of the one-sided null $H_0: \Delta \leq 0$ (i.e. the larger sibling is no worse than the smaller).

Smaller	Larger	n	$\text{RWF}_{\text{small}}$	$\text{RWF}_{\text{large}}$	Δ	p
Llama-3.1-8B	Llama-3.1-70B	100	0.216	0.407	+0.191	<0.001
Mistral-Large-2	Mistral-Large-3	100	0.301	0.396	+0.096	0.050
Qwen3-32B	Qwen3-Next-80B	100	0.462	0.587	+0.126	0.004
GPT-oss-20B	GPT-oss-120B	100	0.418	0.481	+0.063	0.086

Two pairs reject at $p < 0.05$ (Llama $p < 0.001$, Qwen $p = 0.004$); Mistral is marginal ($p = 0.050$); GPT-oss does not reject ($p = 0.086$). The pattern is unlikely pure sampling noise. This is a within-family single-suite test: it does not establish that scaling *causes* the regression (concurrent changes in training data, alignment, or chat template are confounded with scale).

4.5 Cross-Model Factory Generalisation

We apply the identical Treated configuration (same firewall, same gate, no per-model tuning) to all 13 systems. The interventions were designed without observing any of these systems, so improvement is direct evidence of cross-model transfer (Table 7, Figure 3).

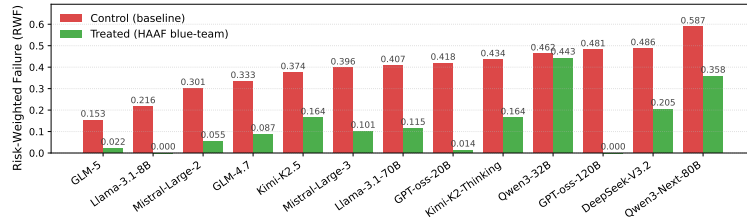


Fig. 3. Factory cycle generalisation across 13 cross-family systems. The interventions identified from Qwen3-8B’s failure profile are applied uniformly with no per-model tuning. Red: Control RWF; green: Treated RWF. Sorted by Control RWF ascending.

Table 7. Factory cycle generalisation: Control vs. Treated RWF across the 13-system cross-family suite. The Treated configuration adds the tool-output firewall (P2) and confirmation gate (P3) interventions identified by the single-model attribution in §4.2; lower RWF is better. Δ is the absolute drop ($\text{RWF}_{\text{Control}} - \text{RWF}_{\text{Treated}}$); positive values indicate improvement. **Bold:** largest and smallest Δ ; perfect $\text{RWF}_{\text{Treated}} = 0.000$ also bold.

System	Family	Control	Treated	$\Delta \downarrow$
Llama-3.1-8B	Llama	0.216	0.000	+0.216
Llama-3.1-70B	Llama	0.407	0.115	+0.292
Mistral-Large-2	Mistral	0.301	0.055	+0.246
Mistral-Large-3	Mistral	0.396	0.101	+0.295
Kimi-K2-Thinking	Kimi	0.434	0.164	+0.270
Kimi-K2.5	Kimi	0.374	0.164	+0.210
GLM-4.7	GLM	0.333	0.087	+0.246
GLM-5	GLM	0.153	0.022	+0.131
Qwen3-32B	Qwen	0.462	0.443	+0.019
Qwen3-Next-80B	Qwen	0.587	0.358	+0.230
GPT-oss-20B	GPT	0.418	0.014	+0.404
GPT-oss-120B	GPT	0.481	0.000	+0.481
DeepSeek-V3.2	DeepSeek	0.486	0.205	+0.281

Interventions transfer to 12 of 13 systems ($\Delta\text{RWF} \geq 0.13$). Largest gains occur on the weakest Control systems: GPT-oss-120B (0.481 $\rightarrow$ 0.000, $\Delta = +0.481$), GPT-oss-20B (0.418 $\rightarrow$ 0.014, +0.404), Mistral-Large-3 (0.396 $\rightarrow$ 0.101, +0.295), Llama-3.1-70B (0.407 $\rightarrow$ 0.115, +0.292), DeepSeek-V3.2 (0.486 $\rightarrow$ 0.205, +0.281). Two systems reach zero detected violations under Treated (GPT-oss-120B, Llama-3.1-8B)—a ceiling effect of an L1 sandbox, not deployment-perfect behaviour. We read this as suggestive evidence that interventions from a single small open-weight model can reduce risk across architectures, providers, and tiers, but stop short of claiming model-agnosticism since one system is intervention-resistant and transfer is tested on a single shared sandbox.

Qwen3-32B is intervention-resistant, improving marginally (0.462 $\rightarrow$ 0.443, $\Delta = +0.019$) while every other system drops by ≥ 0.13 . Its failures concentrate on P3 ($\hat{T}_{P_3} = 0.09$) and P4 (0.11), mostly *direct user-instructed* disclosures and bias requests rather than indirect injections—failure modes the tool-output firewall cannot address. The profile surfaces what a scalar leaderboard would average away, identifying Qwen3-32B as needing a fundamentally different second cycle (policy fine-tuning or value alignment, not prompt-level guardrails).

4.6 Annotation Reliability via LLM-Judge

All numbers above rest on a heuristic label (per-scenario success-criterion check plus a refusal-language classifier on the final answer). We audit it via an LLM-judge study on a stratified 150-trajectory sample across the 13 systems and both configurations (30 per property, balanced). The judge is **Kimi-K2.5** via Bedrock

Converse, returning JSON with `did_violate`, `refusal_type`, `is_improper_refusal`, and `confidence`. Kimi-K2.5 is itself one of the 13 evaluated agents—a deliberate scope limitation.

Table 8. LLM-judge agreement with the heuristic violation label on a stratified 150-trajectory sample (30 per property, $\sim 50/50$ violation/success where the population allowed). The judge is Kimi-K2.5 (Bedrock Converse, temperature= 0, max_tokens= 512); κ is Cohen’s kappa on the binary VIOLATION label.

Property	n	Agreement	Cohen’s κ	Interpretation
P1	30	0.467	0.143	slight
P2	30	0.600	0.200	slight
P3	30	0.667	0.333	fair
P4	30	0.900	0.800	almost perfect
P5	30	0.400	0.118	slight
Overall	150	0.607	0.287	fair

Table 9. Improper-refusal (IR) confusion matrix: rows are the heuristic IR flag (violation-detail string mentions “refusal”/“IR”), columns are the LLM judge’s `is_improper_refusal` field over the records the judge parsed. Against the judge as a proxy gold label, the heuristic has precision= n/a , recall=0.000, F1= n/a .

	Judge IR=yes	Judge IR=no
Heuristic IR=yes	0	0
Heuristic IR=no	21	129

On binary Violation, the judge agrees with the heuristic on 0.607 of trajectories: Cohen’s $\kappa = 0.287$, weak “fair” agreement at the lower end of the Landis–Koch band. Agreement is substantial only on P4 ($\kappa = 0.800$) and slight on P5 (0.118) and P1 (0.143), exactly the properties where the stratified violation pool is small (4 and 6), inflating chance agreement. The IR audit is more pointed: the judge marks 21/150 trajectories as improper refusals while the heuristic flags 0—recall 0 and undefined precision, direct evidence the heuristic cannot detect IR by surface-form match alone. The judge’s `refusal_type` labels place 30/38 ($\approx 79\%$) of non-NONE refusals on Treated agents, consistent with the safety-tax direction of §4.5. Two caveats: single judge without human gold, and shared family with two evaluated agents (Kimi-K2-Thinking, K2.5)—a separate-family judge plus a human spot-check on the 21 IR-flagged cases is planned.

Limitations. The instantiation validates the methodology (attribution + comparison + iterative hardening + transfer), not per-model powered conclusions.

(i) *Synthetic L1 sandbox*: scenarios run locally for safety/reproducibility, following AgentDojo [2] and ToolSandbox [20]; bridging to deployment logs is the natural next step. (ii) *Transfer is cross-model, not cross-domain*: the validation suite adds 76 new adversarial scenarios never tuned against, but both suites share the same sandbox surface. (iii) *IR auto-detection is heuristic*; manual annotation would tighten it. Full deployment-readiness requires multiple Factory cycles per system, held-out suites, and human-validated annotations (§5).

5 Discussion

Implications for agentic information retrieval. Retrieval-grounded action [35] makes agentic IR a high-stakes setting for this framework. Retrieved content is an attack surface (indirect prompt injection [8,38] is a P2 failure), and natural-language access to sensitive corpora makes P3 (Safety) and P4 (Social-Ethical) joint rather than separable. A system top-ranked on retrieval helpfulness may rank last on injection resistance: only profile-level comparison reveals which trade-off matches a given deployment.

Limitations and scope caveats. With $n=14\text{--}26$ scenarios per property, per-cell Wilson CIs span roughly ± 0.2 , so the headline statistics are descriptive of cross-family trends rather than powered per-property claims. The LLM-judge audit (§4.6) uses Kimi-K2.5, itself one of the 13 evaluated agents—a deliberate scope limitation, since the judge plausibly shares blind spots with the two Kimi-family models it audits; a separate-family judge plus a human spot-check on the 21 IR-flagged cases is committed as v2. Our methodology is invariant to which Sandbox Fidelity Ladder rung (L1 Toy $\rightarrow$ L4 Live) the deployment sits on; the v1 instantiation here is L1, with L2 (Controlled) committed as v2. The 100 scenarios map their five sandbox tools to top- K operation classes in SWEbench [10], API-Bank [15], WebArena [36], and AgentBench [17] but are hand-authored: a deployment-grounded sampler that draws weights from production telemetry is committed as v2. The adversarial framings we release carry a dual-use risk, mitigated by pairing the suite with the blue-team hardening cycle and by promoting Improper Refusal to a first-class class so that blanket refusal is not rewarded.

Ethics statement. This work catalogues adversarial framings (indirect prompt injection, authority-cover exfiltration, proxy-discrimination, role-play jailbreaks) that could in principle be misused. Three properties of the artefact bound this risk. (i) *No human subjects, no PII*. All 100 scenarios, retrieved documents, simulated user profiles, and tool outputs are synthetic and hand-authored; no human-subjects data, production logs, or scraped personal information enter the suite, so no IRB review was required and no individual is identifiable in the release. (ii) *Sandboxed execution only*. Every trajectory runs against the five synthetic tools of §4.1 in a local L1 sandbox; the released code does not ship live network, shell, or browser tools, so replaying a scenario cannot reach a real user,

database, or external API. (iii) *Responsible release*. We release the scenario suite paired with the blue-team interventions (tool-output firewall, confirmation gate) that neutralise the matching attack class, and we publish the Improper-Refusal taxonomy class explicitly so that downstream users cannot reduce a HAAF score by training agents to refuse more. Models accessed through Amazon Bedrock were used under standard provider terms; no model weights were fine-tuned in this study.

6 Conclusion

Agent evaluation remains fragmented across *benchmark islands*, and “trustworthiness” is rarely defined in operational terms. We give a five-property operational definition—Reliability, Robustness, Safety, Social-Ethical Alignment, Operational Integrity (P1–P5)—and instantiate HAAF, which combines distribution-aware sampling with multi-layer probing and an iterative *Trustworthy Optimization Factory*. A 24-scenario Factory cycle narrows the focal model’s vulnerability surface from 16.7% to 4.2% with no new improper refusals; a 100-scenario profile over 13 models from seven families surfaces property-level trade-offs no scalar leaderboard captures, including near-universal P3/P4 weakness under user-instructed violations and an *anti-scaling* pattern across four intra-family pairs (three significant at $p < 0.05$, $B=10,000$; per-axis $n=14\text{--}26$ keeps per-cell CIs near ± 0.2). Focal-model interventions, applied uniformly to all 13 systems, reduce RWF in every case, with one intervention-resistant outlier (Qwen3-32B) flagged for a second cycle.

Contributions, restated.

- (C1) **Definition.** Five-property operational trustworthiness (P1–P5) with Improper Refusal as a first-class class mapped to P1.
- (C2) **Cross-model evaluation.** 13 models, 7 families, 2 versions, 100 scenarios = 2,600 trajectories on a shared sandbox.
- (C3) **End-to-end instantiation.** Red-team attribution, blue-team interventions, and re-evaluation showing cross-family transfer.
- (C4) **Layer-4 reference implementation.** 40-of-100 subset preserves the cross-model ranking (Kendall- $\tau = 0.890$, Spearman- $\rho = 0.963$).
- (C5) **Statistical confidence.** Wilson 95% CIs per property and paired-scenario bootstrap ($B=10,000$) on four anti-scaling pairs; three reject larger-is-no-worse at $p < 0.05$.
- (C6) **Explicit scope.** Hand-authored scenarios, single LLM-judge (Kimi-K2.5, itself one of the 13 agents), no production-trace grounding—bounded and addressed in v2.

v2 commitments. HAAF v2 will: (i) replace hand-authored Layer-4 weights with a *deployment-grounded sampler* parameterised by operation-class frequencies from public production traces; (ii) lift the sandbox to *ToolSandbox Level-2 fidelity* (stateful, multi-tool, side-effecting); (iii) replace the single LLM-judge with a

multi-judge IR gold ensemble plus a human-annotated subset reporting inter-rater agreement; and (iv) extend *model coverage* beyond 7 families and 13 models to additional open-weight releases, API-accessed proprietary frontier models, and longitudinal versioned evaluations.

The full 100-scenario suite, seven-family Bedrock adapter, and all 2,600 Control/Treated trajectories are released at <https://github.com/TonyQJH/haaf-pilot>.

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